

Separation of variable

Calculus II (Exam)

1) find $\frac{dy}{dx}$ $x^2 + y^2 = k \longrightarrow 2x + 2y \frac{dy}{dx} = 0$

$$2(x + y \frac{dy}{dx}) = 0$$

$$x + y \frac{dy}{dx} = 0 \longrightarrow \frac{dy}{dx} = -\frac{x}{y}$$

$$y(g(x))^2 = 2y(g(x)) \cdot y'(x)$$

Now in Different Equat we do the oposite stuff, we start with the solution

1° $x + y \frac{dy}{dx} = 0$

1) Formed / 2) Diagnostic / 3) Separate

$$y \cdot \frac{dy}{dx} = -x \quad \checkmark$$

4) Integrate

$$\int y \cdot \frac{dy}{dx} \cdot dx = \int -x \, dx$$

at least 1 C in the equation

$$\int y \, dy = \int -x \, dx \longrightarrow \frac{y^2}{2} = -\frac{x^2}{2} + C$$

5) Solve $\rightarrow$ General F(x)

$$\frac{y^2}{2} = -\frac{x^2}{2} + C \longrightarrow y^2 = -x^2 + C$$

$$y = \pm \sqrt{-x^2 + C}$$

2° $\frac{dy}{dx} = y^2 - 4$; $y(0) = -2$

$$\frac{dy}{dx} = (y^2 - 4) \cdot 1 \, dx$$

$$\frac{dy}{dx} = \frac{M(x)}{N(y)}$$

$$\left(\frac{1}{y^2 - 4}\right) \frac{dy}{dx} = 0 \longrightarrow N(y) \frac{dy}{dx} = M(x)$$

$$M(x) = 1$$

$$N(y) = \frac{1}{y^2 - 4}$$

$$H_1(x) = \int 1 \, dx = x$$

$$H_2(y) = \int \frac{1}{y^2 - 4} \, dy$$

Little Problem

Partial Fraction decomposition

$$\int \frac{1}{y^2-4} dy \longrightarrow \frac{1}{(y+2)(y-2)}$$

why? Cause there's not rule to solve it directly
 $\int \frac{1}{u} du = \ln|u| + C$
 only works when u is a linear expression.

3) Set up the decomposition

$$\frac{1}{(y+2)(y-2)} = \frac{A}{(y+2)} + \frac{B}{(y-2)}$$

4) $1 = \left[\frac{A}{(y+2)} + \frac{B}{(y-2)} \right] (y+2)(y-2)$

$$1 = A(y-2) + B(y+2)$$

5) solve for A & B

let $y = -2$

$$1 = A(-2-2) + B(-2+2)$$

$$A = -\frac{1}{4}$$

let $y = 2$

$$1 = A(2-2) + B(2+2)$$

$$B = \frac{1}{4}$$

6)
$$\frac{1}{(y+2)(y-2)} = \frac{-\frac{1}{4}}{(y+2)} + \frac{\frac{1}{4}}{(y-2)}$$

$$= \frac{-1}{4y+8} + \frac{1}{4y-8}$$

• Little Problem:

$$H_2(y) = \int \frac{1}{y^2-4} dy = \int \frac{-1}{4y+8} + \frac{1}{4y-8} dy \longrightarrow \int \frac{-1}{4y+8} dy + \int \frac{1}{4y-8} dy$$

1) $\int \frac{-1}{4y+8} dy \rightarrow u = 4y+8 \rightarrow \frac{-1}{4} \int \frac{1}{u} du = -\frac{1}{4} \ln(u) = -\frac{1}{4} \ln|4y+8|$
 $du = 4dy$
 $dy = \frac{du}{4}$
 Internal derivative

2) $\int \frac{1}{4y-8} dy \rightarrow u = 4y-8 \rightarrow \frac{1}{4} \int \frac{1}{u} du = \frac{1}{4} \ln(u) = \frac{1}{4} \ln|4y-8|$
 $du = 4dy$
 $dy = \frac{du}{4}$

$$H_2(y) = -\frac{1}{4} \ln(4y+8) + \frac{1}{4} \ln(4y-8)$$

Continuing with the Problem

$$\frac{dy}{dx} = (y^2 - 4) \cdot 1 \cdot dx \quad \frac{dy}{dx} = \frac{M(x)}{N(y)} \quad \int N(y) \cdot \frac{dy}{dx} dx = \int M(x) dx$$

$$\left(\frac{1}{y^2-4}\right) \frac{dy}{dx} = 0 \rightarrow N(y) \frac{dy}{dx} = M(x)$$

$$M(x) = 1 \quad H_1(x) = \int 1 dx = x$$

$$N(y) = \frac{1}{y^2-4} \quad H_2(y) = -\frac{1}{4} \ln(4y+8) + \frac{1}{4} \ln(4y-8)$$

$$\frac{1}{4} (\ln(4y-8) - \ln(4y+8))$$

$$\int N(y) \cdot \frac{dy}{dx} dx = \int M(x) dx$$

$$\frac{1}{4} \cdot (\ln(4y-8) - \ln(4y+8)) = x + C$$

$$\ln(a) - \ln(b) = \ln\left(\frac{a}{b}\right)$$

$$\ln(4y-8) - \ln(4y+8) = 4x + C$$

$$\ln\left(\frac{4y-8}{4y+8}\right) = 4x + C$$

$$e^{\ln\left(\frac{4y-8}{4y+8}\right)} = e^{4x+C}$$

$$\frac{4y-8}{4y+8} = ke^{4x+C}$$

$$4y-8 = ke^{4x+C} (4y+8)$$

$$4y-8 = y ke^{4x+C} + ke^{4x+C}$$

$$4y = y ke^{4x+C} + ke^{4x+C} + 8$$

$$4y - y ke^{4x+C} = ke^{4x+C} + 8$$

$$y(4 - ke^{4x+C}) = ke^{4x+C} + 8$$

$$y = \frac{ke^{4x+C} + 8}{4 - ke^{4x+C}}$$

$$y = \frac{8(ke^{4x+C} + 1)}{4(ke^{4x+C} - 1)}$$

$$y = \frac{2(ke^{4x+C})}{ke^{4x+C} - 1}$$